Arnav Parekar

Gmail | LinkedIn | Github | Leetcode

Summary

CS student specializing in cybersecurity and cloud infrastructure. Experienced in Python scripting, cloud platforms and security monitoring through OT security internships and projects. Interested in building reliable, scalable systems and improving security visibility through observability, automation, and incident response.

Education

Vellore Institute of Technology, Chennai — CGPA : 9.16/10
Bachelor of Technology in Computer Science — *July 2023 – May 2027*

Technical Skills

Vulnerability Assessment: Nmap, Nessus, OpenVAS, Network Scanning
Network Analysis: Wireshark, tcpdump, Network Protocol Analysis, Traffic Inspection
Cloud & Deployment: Google Cloud Platform, Microsoft Azure, Docker, CI/CD
Programming & Scripting: Python, Java, C/C++, JavaScript, Bash Scripting, PowerShell
Security Frameworks: IEC62443, NIST, ISO 27001, MITRE ATT&CK Framework, Risk Assessment

Experience

OT Cybersecurity Intern | *OSICS Technologies, Pune* — June 2025 -July 2025

- Monitored ICS network traffic and analyzed operational behavior across containerized SCADA environments and drafted **risk assessment** report for an oil and gas setup.
- Gained hands-on experience with OT network architectures and ICS protocols **(Modbus, DNP3, OPC UA)** through the **LabShock** tool, analyzing traffic and **simulating attack scenarios** in a containerized SCADA setup.

Mathematics Club Technical Lead | *VIT Chennai* — July 2024 - June 2025

- **Leading** technical initiatives, organizing workshops, and **mentoring** students in advanced mathematical problem-solving thus reaching out to more than 300 students.

Achievements

Bloom Energy Hackfest Winner | *National* — April 2026

- Developed an cybersecurity solution with an AI first approach for IDS and secured an internship.

IEEE YESIST 26 Finalists | *International* — October 2026

- Won the prelims and got selected as global finalists for developing intelligent travel safety solution.

Projects

ICS-Watchdog | *Modbus, Packet Analysis, MITRE ATT&CK, Docker, Python* — Link | June 2026

- Containerised Modbus/TCP threat detection lab (Docker) simulating PLC master/slave communication, injecting MITRE ATT&CK for **ICS attack scenarios** (reconnaissance, coil injection, replay), and **detecting** them with a stateful rule engine producing JSON/HTML **audit reports**.
- Implemented **8 detection rules** covering MITRE ATT&CK for ICS techniques T0846, T0855, T0856, T0884 - each rule maps directly to a **protocol-level Modbus anomaly** (function code abuse, register boundary violations, replay patterns).

ICS-ZoneAudit | *Nmap, Audit and Compliance, IEC62443, NIST, Docker, Python* — Link | June 2026

- Python CLI that parses Nmap XML scans of industrial networks, **classifies assets** into IEC 62443 security zones (Enterprise/DMZ/Control/Field Device), **detects** cross-zone conduit violations, and scores each asset against a **15-point hardening checklist** derived from IEC 62443-3-3 and NIST SP 800-82.
- Generates **structured compliance reports** (JSON + HTML) including a zone topology diagram and per-asset remediation guidance - operationalising IEC 62443-3-2 **zone/conduit modelling** as executable code.

Patents

System for Monitoring Vehicle Speed in Traffic Environments and Method of Speed Estimation | *Patent Filed & Published* | *Application No: 202541130222*